



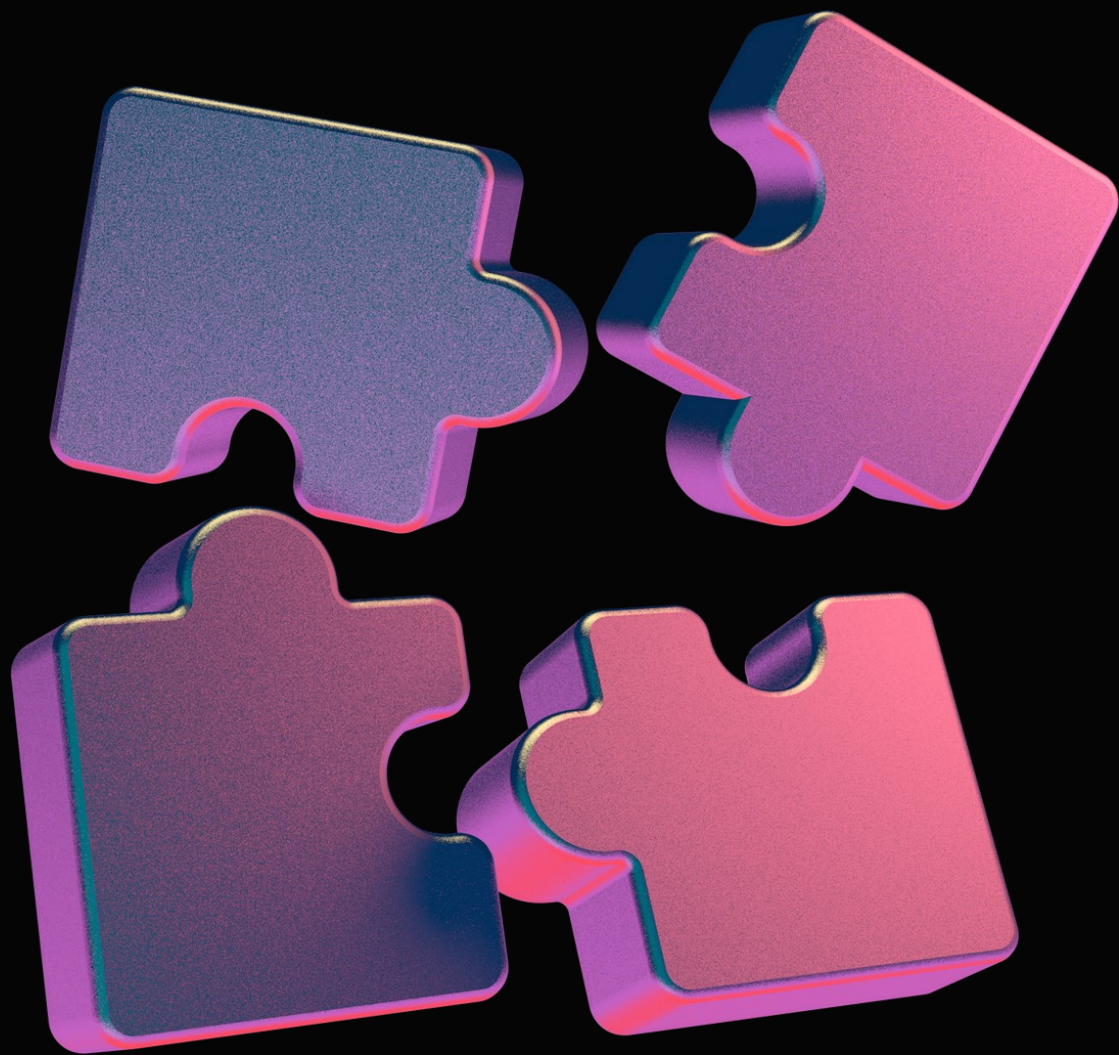
AITracker

Project Presentation

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Scanner Device Identification from Scanned Documents



Scanned documents contain hidden device-specific patterns invisible to the human eye—subtle artifacts and noise signatures unique to each scanner model.

Our Objective: Build an intelligent system that can accurately predict the scanner brand from scanned images using machine learning and deep learning techniques.

Dataset Collection & Understanding

Dataset Composition

We assembled a comprehensive collection of scanned document images from three major scanner manufacturers: Canon, HP, and Epson. Each brand exhibits distinctive scanning characteristics that serve as forensic fingerprints.

- 1000+ scanned document images
- Multiple resolutions: 150 DPI and 300 DPI
- High-quality TIFF format for maximum detail
- Carefully labeled dataset with brand annotations



Image Preprocessing Pipeline

01

Grayscale Conversion

Converted color images to grayscale to focus on texture and noise patterns rather than color variations

02

Image Resizing

Standardized all images to a fixed dimension to ensure consistent input for model training

03

Pixel Normalization

Scaled pixel values to a standard range, improving model convergence and training stability

04

TIFF Handling

Implemented specialized processing for high-resolution TIFF files to preserve forensic details

05

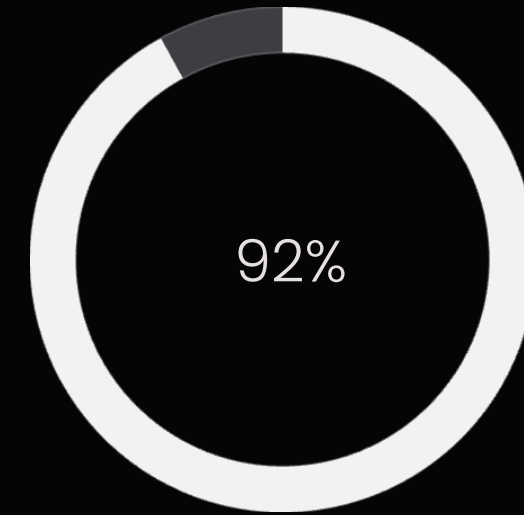
Quality Assurance

Validated preprocessing consistency across entire dataset to eliminate pipeline variability

Feature Extraction & Classical Machine Learning

Local Binary Pattern Features

We extracted Local Binary Pattern (LBP) features—a powerful texture descriptor that captures the micro-patterns and noise characteristics unique to each scanner. LBP excels at detecting subtle variations in pixel neighborhoods that distinguish scanner signatures.



SVM Accuracy
Best performance on 300 DPI dataset

Classical ML Models Tested

- Support Vector Machine (SVM)
- Random Forest Classifier
- K-Nearest Neighbors (KNN)
- Logistic Regression



Key Finding: Performance degraded when mixing 150 and 300 DPI images, revealing that classical ML models are highly sensitive to resolution changes. This limitation motivated our exploration of deep learning approaches.



Deep Learning Approach: From Scratch to Transfer Learning

1

CNN from Scratch

Initial attempt with custom convolutional neural network revealed severe underfitting due to limited dataset size. Accuracy plateaued at only 40-55%, indicating insufficient training data.

2

Transfer Learning

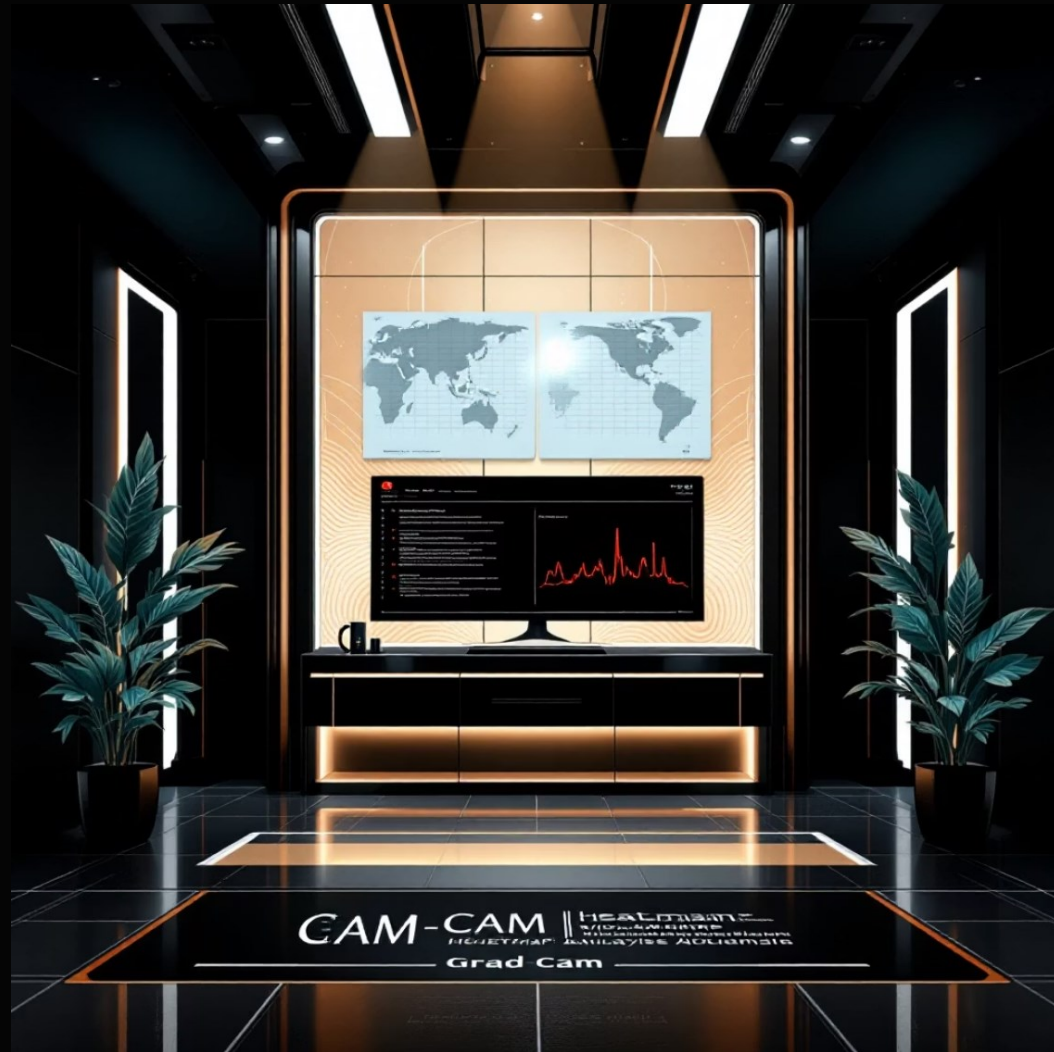
Leveraged pre-trained MobileNetV2 and EfficientNet architectures, fine-tuning them on our scanner dataset. Applied extensive data augmentation to artificially expand training samples.

3

Performance Breakthrough

Achieved dramatic improvement with 82% training accuracy and 84%+ validation/testing accuracy. Transfer learning effectively addressed the small dataset challenge.

Model Explainability: Understanding CNN Decisions



Grad-CAM Visualization

To understand what our CNN models actually learned, we employed Gradient-weighted Class Activation Mapping (Grad-CAM), a powerful technique that highlights which image regions most influence predictions.

Critical Observations

- Model attention concentrated on background areas and noise patterns
- Minimal focus on text content itself
- Strong activation in margin regions and uniform backgrounds

These visualizations confirm our hypothesis: the model successfully learns scanner-specific artifacts—the subtle noise signatures and textural patterns each device imprints—rather than document content. This validates the forensic soundness of our approach.

Deployment & Key Takeaways

Streamlit Web Interface

Deployed an intuitive web application where users can upload scanned documents and receive instant scanner brand predictions. The interface maintains the exact preprocessing pipeline used during training, ensuring prediction consistency.

Project Conclusions

- **SVM + LBP:** Optimal for single-DPI scenarios with 92% accuracy
- **CNN Models:** Superior generalization across different DPI settings
- **Transfer Learning:** Essential for small forensic datasets

Complete ML Pipeline

Demonstrated end-to-end forensic machine learning: from data collection through model explainability to production deployment

Forensic Applications

Practical tool for document authentication, fraud detection, and legal investigations requiring scanner source attribution

Future Directions

Expand to more scanner brands, explore printer identification, and integrate with document verification systems

